

Roll No.

TEC-304

B. TECH. (ECE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022

SIGNALS AND SYSTEMS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Determine whether or not each of the following signals is periodic. If a signal is periodic, determine its fundamental period : (CO1)

(i) $x(t) = \cos(t) + \sin(\sqrt{2}t)$

(ii) $x[n] = e^{\frac{j2\pi n}{3}} + e^{\frac{j3\pi n}{4}}$

OR

- (b) Determine whether the following signals are energy signals, power signals, or neither : (CO1)

(i) $x(t) = e^{-at}u(t), a > 0$

(ii) $x(t) = tu(t)$.

2. (a) Determine the even and odd components of the following signals : (CO1)

(i) $x(t) = \cos(t) + \sin(t) + (\sin)^*(\cos(t))$

(ii) $x[n] = u[n]$.

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OR

(b) Evaluate the following :

(CO1)

(i) $\sin t \times \delta(t)$

(ii) $\int_{-1}^1 (3t^2 + 1) \delta(t) dt$

(iii) $\int_1^2 (3t^2 + 1) \delta(t) dt$

3. (a) Determine the power and the rms value of the given signals : (CO1)

(i) $x(t) = 10 \cos\left(50t + \frac{2\pi}{3}\right)$

(ii) $x[n] = 2e^{j3n}$

OR

(b) Define continuous time unit step and continuous time unit impulse functions. (CO1)

4. (a) What is LTI system and its impulse response ? Explain, how the impulse response is useful in finding out the LTI system output due to any arbitrary input signal. (CO1)

OR

(b) Check whether the given system $y(t) = \text{odd}\{x(t)\}$ is : (CO1)

(i) Linear or nonlinear

(ii) Time invariant or time variant

5. (a) What is BIBO criterion for stability of a system ? (CO2)

(3)

OR

- (b) Determine whether or not each of the following systems is invertible. If the system is invertible, give the inverse system : (CO2)

(i) $y(t) = x^2(t)$

(ii) $y[n] = x[n] \times x[n - 2]$